RVI to NDVI Model Report

Synthetic Optical Vegetation Index Generation using Radar Polarizations

This technical report provides a detailed breakdown of the machine learning model designed to translate Sentinel-1 Synthetic Aperture Radar (SAR) based **Radar Vegetation Index (RVI)** into Sentinel-2 **Normalized Difference Vegetation Index (NDVI)**. Utilizing physical coordinates, seasonal parameters, dynamic world probabilities, and meteorological variables, the model provides cloud-free synthetic NDVI estimations to support continuous parcel-level agricultural audits.

EVALUATION METHOD

Group-based Train-Test Split (80/20)
Grouped by Parcel ID (task_id)

TRAINING COHORT

244 Train Parcels (18,289 rows)
62 Unseen Test Parcels (4,810 rows)

MODEL TYPE

Random Forest Regressor
June 2026

1. Dataset Alignment & Model Performance

PERFORMANCE SUMMARY

Due to clouds obstructing optical bands, creating a synthetic proxy of NDVI from Sentinel-1 radar bands allows for continuous agricultural health monitoring. To train the translator model, Sentinel-2 optical scenes were paired with their nearest Sentinel-1 observations within a **3-day matching window**, producing a high-quality dataset of 23,099 aligned observations.

Operational Validation: Spatial Generalization Check

By splitting training and testing data at the parcel level (Group-based Split), the model was evaluated on 62 parcels that it had never seen during training. This simulates the exact production environment where the model is requested to generate predictions for entirely new geographical boundaries.

Core Model Metrics

Data Partition	Coefficient of Determination (R^2)	Root Mean Squared Error (RMSE)	Mean Absolute Error (MAE)
Train Set (244 Parcels)	0.9686	0.0442	0.0369
Test Set (Unseen Parcels) (62 Parcels)	**0.8975**	**0.0919**	**0.0634**

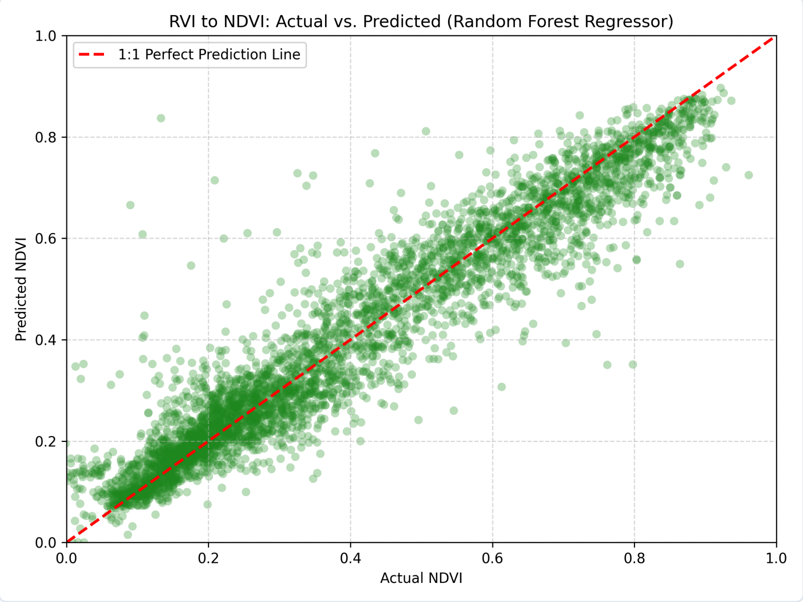

Figure 1: RVI to NDVI actual vs. predicted values on unseen test parcels.

2. Drivers of Vegetation Reconstruction

FEATURE IMPORTANCE & LULC CONTEXT

Feature importance analysis reveals that the target's current radar value (`raw_RVI`) and its temporal dynamics (lags/leads representing recent trends) serve as the dominant indicators. Local meteorological inputs (rainfall, temperature) and dynamic land cover distributions successfully constrain regional variations in signal response.

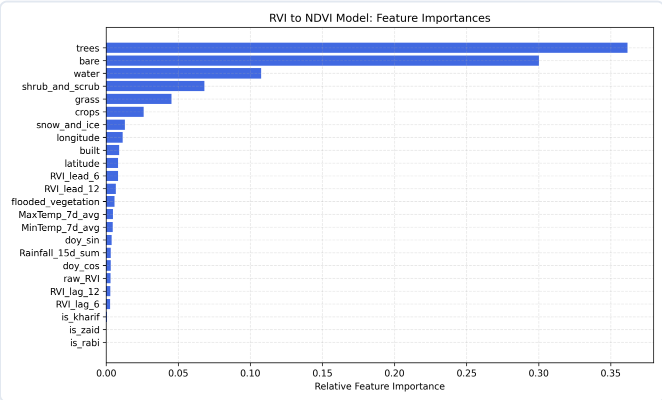

Figure 2: Relative feature importances in Random Forest regressor.

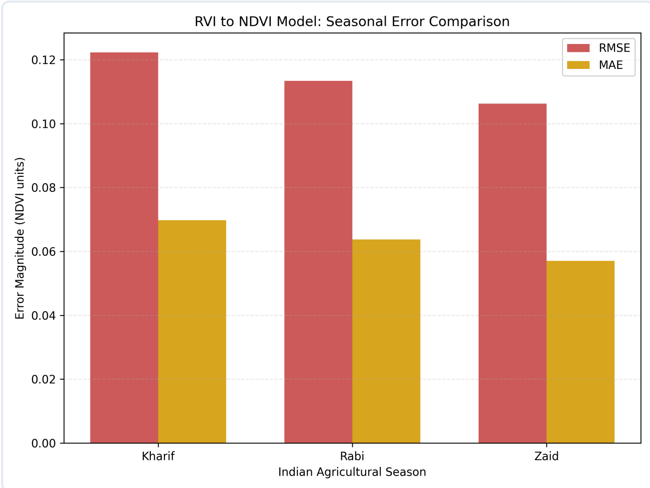

Figure 3: Seasonal error magnitude comparison (RMSE vs MAE).

Temporal Tracking Case Study

The reconstructed NDVI curve (orange) tracks the optical NDVI signature (green) over the full course of multiple seasonal crop cycles. This highlights that the model successfully filters out sensor speckle while maintaining sensitive response times to crop emergence, peak greening, and harvest events.

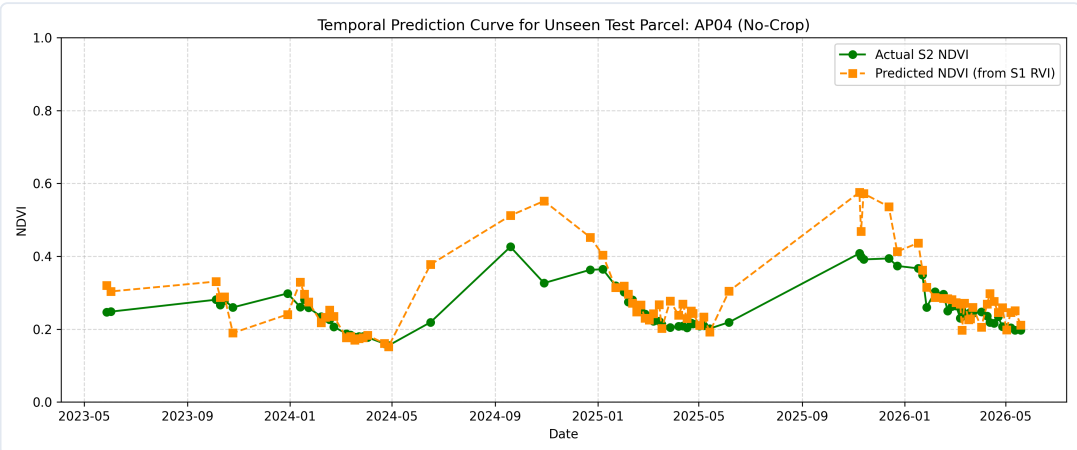

Figure 4: Synthetic NDVI curve predictions vs actual Sentinel-2 optical NDVI on an unseen test parcel.